

Robin CR

Chennai, Tamil Nadu

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Personal Profile

4+ years experienced Embedded Software Developer with proven success in building algorithms. Highly adept at C++ application development, firmware development, ADAS, Design Patterns and multi-threading. Passionate engineer with the ability to apply techniques and algorithm development to solve real world problem.

Work Experience

Automotive Robotics India (ARi)

Senior Software Engineer

Chennai, Tamil Nadu

January 2025 – Present

Client: CATERPILLAR (Onsite)

LIDAR Perception for Surface Hauling truck

- Playback tool developer.
- Rangeimage to point cloud converter for various lidars.
- Amenders for lidar logs.
- Parsers for ROS node communication.
- Developed APIs to work on multi-threaded application using modern C++.
- Integrated third-party libraries (Point Cloud) into C++ project.
- Publisher and subscriber channels in ROS2 Noetic modules.

Bosch Global Software Technologies

Embedded Software Engineer

Coimbatore, Tamil Nadu

August 2022 – December 2024

Interior Monitoring System (IMS) Project

- Developed algorithms for driver drowsiness module in C++ following design patterns and clean code.
- Able to handle real world bugs which require proper strategies, Problem Solving and analytical skills.
- Developed internal tools for simulating driver behaviors for debugging the issues without hardware.
- A proper research and contributor for technical problems raised by client.
- Modules implementations such as interfaces, precondition handlers, error handlers, schedulers etc.
- Successfully addressed and resolved Level 8 and 9 QAC++ warnings in the codebase, adhering to MISRA coding standards.

Instrument Cluster Project

- Worked in all phases of Agile Scrum methodology (Requirement - Design - Coding - Testing).
- Issue reproduction based on scenarios (Bugs and Change Requests).
- Functionality testing based on all dependent modules.
- Writing unit tests and performing testing to ensure software functionality.
- Implemented an algorithm that will modify the fixed parameters in RTE layer into a configurable in application layer.
- Base level knowledge on Diagnostics, Bootloader and RTE layer is gained.
- Cmake integration to the modules.

Gadgeon Systems Pvt Ltd

Embedded Software Developer

Kochi, Kerala

July 2021 - August 2022

- Developed codecs for different sensors using C programming language in a Linux OS environment. (Gateway Programming)
- Developed custom protocols to communicate with different sensors in embedded C.
- Platform dependent and independent code development on different ARM based microcontrollers.
- Interfaced accelerometer, gyroscope, and GPS sensors (Bare metal programming).

Projects

Calendar and Clock

Self

- A project which addresses all the problems related to calendar and clock.

CPU-GPU Selector

- Implemented a system that decides the model training to be done on CPU, GPU, or GPU on cloud.

Skills

- **Programming language:** Proficient in Embedded C, C++
- **Microcontrollers:** ARM Cortex M3 LPC1768, ESP8266
- **Protocols:** UART, I2C, SPI, CAN
- **Programming IDE:** Visual Studio Code, VS Professional, Eclipse IDE
- **Requirement Software:** IBM Rational DOORS
- **Design:** Enterprise Architecture, PlantUML
- **Debugger:** Valgrind, GDB, Lauterbach Trace32
- **Automotive testing tools:** CANoe
- **Project Manager:** Gerrit, RTC workflow manager, JIRA
- **Unit Testing:** Gtest, Excel UT automation, BOOST
- **Version control system:** Git, RTC, BitBucket
- **OS:** Ubuntu

Competencies

- **Data Structures**
Array, Linked list, STL library.
- **Embedded C programming**
State machine, Bit wise programming, handshaking mechanisms.
- **C++ Application development**
Move Semantics, Smart Pointers, Templates, Functional Programming, Atomics.

Certifications

Embedded System Professional

Translot, Kochi

- Solid foundation on basics of embedded C programming.
- Completed the projects based on EEPROM and display systems.
- Working knowledge on resource utilization, shared memory, and schedulers.
- Proper understanding of Linux Fundamentals (System calls, Booting, file system) is gained.

Professional C++ and Clean code

Bosch, Coimbatore

- Leveraged the OOPs and SOLID principles with examples.
- Discussed common pitfalls in C++ for embedded software development.
- Covered Design Patterns explaining their roles in code reusability and scalability.
- Clean coding guidelines and commons practices were addressed.

Education

Rajagiri School of Engineering and Technology

Bachelor of Technology in ECE

Completed with a GPA of 8.2

Kochi, Kerala

Aug 2017 - Jul 2021